

## Homework 1

### PROBLEM 1

**(a) (i) Isothermal:**  $T = \text{constant}$ .

$$T_{2,\text{iso}} = T_1 = \boxed{300 \text{ K}}, \quad pV = nR_u T_1 = \text{constant}, \quad \frac{V_{2,\text{iso}}}{V_1} = \frac{p_1}{p_2} = \frac{1}{8}.$$

For a calorically perfect gas,  $\Delta E = n\bar{c}_v(T_2 - T_1) = 0$ . Thus the first law gives  $0 = Q - W_{\text{by}}$ . The reversible boundary work is

$$\begin{aligned} W_{\text{by,iso}} &= \int_{V_1}^{V_{2,\text{iso}}} p dV = nR_u T_1 \int_{V_1}^{V_{2,\text{iso}}} \frac{dV}{V} \\ &= nR_u T_1 \ln \frac{V_{2,\text{iso}}}{V_1} = -(1)(8.31446)(300) \ln 8 = \boxed{-5.187 \text{ kJ}}. \end{aligned}$$

**(ii) Reversible adiabatic:**  $Q = 0$  and  $S_{\text{gen}} = 0$ , so  $\Delta S = 0$ .

$$\begin{aligned} pV^\gamma &= \text{constant}, & \frac{T_{2,\text{ad}}}{T_1} &= \left(\frac{p_2}{p_1}\right)^{(\gamma-1)/\gamma}, \\ T_{2,\text{ad}} &= 300(8)^{0.4/1.4} = \boxed{543.4 \text{ K}}. \end{aligned}$$

$$\begin{aligned} \bar{c}_p &= \bar{c}_v + R_u, & \gamma &= \frac{\bar{c}_p}{\bar{c}_v}, \\ (\gamma - 1)\bar{c}_v &= R_u, & \bar{c}_v &= \frac{R_u}{\gamma - 1} = 20.7862 \text{ J}/(\text{mol K}). \end{aligned}$$

Apply the first law with  $Q = 0$ :

$$\begin{aligned} \Delta E &= Q - W_{\text{by,ad}} = -W_{\text{by,ad}}, \\ W_{\text{by,ad}} &= -\Delta E = -m\Delta e = -n\bar{c}_v(T_{2,\text{ad}} - T_1) = \boxed{-5.060 \text{ kJ}}. \end{aligned}$$

**(b)** Use the ideal-gas entropy relation and convert totals to mass-specific values:

$$\begin{aligned} \Delta S_{\text{sys}} &= n \left( \bar{c}_p \ln \frac{T_2}{T_1} - R_u \ln \frac{p_2}{p_1} \right), & \bar{c}_p &= \frac{\gamma R_u}{\gamma - 1}, \\ m = nM_{\text{O}_2} &= (1 \text{ mol})(0.03200 \text{ kg/mol}) = 0.03200 \text{ kg}, & \Delta e &= \frac{\Delta E}{m}, \quad q = \frac{Q}{m}, \quad \Delta s_{\text{sys}} = \frac{\Delta S_{\text{sys}}}{m}. \end{aligned}$$

**(i) Isothermal:**

$$\begin{aligned} \Delta E &= 0, & Q &= W_{\text{by,iso}} = -5.187 \text{ kJ}, \\ \Delta S_{\text{sys}} &= n(\bar{c}_p \ln 1 - R_u \ln 8) = -nR_u \ln 8 = -17.29 \text{ J/K}, \\ \boxed{\Delta e = 0}, & \quad \boxed{q = -162.1 \text{ kJ/kg}}, & \quad \boxed{\Delta s_{\text{sys}} = -0.5403 \text{ kJ}/(\text{kg K})}. \end{aligned}$$

**(ii) Reversible adiabatic:**

$$\begin{aligned} \Delta E &= n\bar{c}_v(T_{2,\text{ad}} - T_1) = +5.060 \text{ kJ}, & Q &= 0, \\ \Delta S_{\text{sys}} &= n \left[ \frac{\gamma R_u}{\gamma - 1} \frac{\gamma - 1}{\gamma} \ln 8 - R_u \ln 8 \right] = 0, \\ \boxed{\Delta e = +158.1 \text{ kJ/kg}}, & \quad \boxed{q = 0}, & \quad \boxed{\Delta s_{\text{sys}} = 0}. \end{aligned}$$

**(c) (i) Isothermal:**

$$Q_{\text{surr}} = -Q = +5.187 \text{ kJ},$$

$$\Delta S_{\text{surr}} = \frac{Q_{\text{surr}}}{300 \text{ K}} = +17.29 \text{ J/K},$$

$$\Delta S_{\text{universe}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = -17.29 + 17.29 = \boxed{0}.$$

**(ii) Reversible adiabatic:**

$$\boxed{\Delta S_{\text{sys}} = \Delta S_{\text{surr}} = \Delta S_{\text{universe}} = 0.}$$

The ideal work source transfers energy without entropy. Both processes have zero entropy generation because they are reversible.

**(d)**

$$W_{\text{in}} = -W_{\text{by}}, \quad \boxed{W_{\text{in,iso}} = 5.187 \text{ kJ}}, \quad \boxed{W_{\text{in,ad}} = 5.060 \text{ kJ}},$$

$$V_{2,\text{iso}} = \frac{nR_u T_1}{p_2} = 3.118 \text{ L}, \quad V_{2,\text{ad}} = \frac{nR_u T_{2,\text{ad}}}{p_2} = 5.648 \text{ L}.$$

For these data, the adiabatic path requires 0.1268 kJ less work input. Its temperature rises, so it reaches 8 bar at a larger volume and stops sooner. The final pressures are the same, but the final states are different. Comparing the paths at the same final volume would give a different work comparison.

## PROBLEM 5

- (a)  $h$ : hot block;  $c$ : cold block. Each block has total heat capacity  $C = 500 \text{ J/K}$ .

$$\begin{aligned} Q_{\text{ext}} = W_{\text{ext}} = 0 &\implies \Delta E_h + \Delta E_c = 0, \\ C(T_f - 500) + C(T_f - 300) &= 0, \\ T_f &= \boxed{400 \text{ K}}. \end{aligned}$$

- (b) Along a reversible auxiliary path,

$$\delta Q_{\text{rev}} = dE = C dT \implies \Delta S = \int_{T_i}^{T_f} \frac{C dT}{T} = C \ln \frac{T_f}{T_i}.$$

$$\Delta S_h = 500 \ln \frac{400}{500} = \boxed{-111.57 \text{ J/K}},$$

$$\Delta S_c = 500 \ln \frac{400}{300} = \boxed{+143.84 \text{ J/K}},$$

$$\Delta S_{\text{universe}} = \Delta S_h + \Delta S_c = \boxed{32.27 \text{ J/K}} = S_{\text{gen}} > 0.$$

- (c) Take heat transferred from the hot block to the cold block as positive.

$$dE_h = -\delta Q, \quad dE_c = \delta Q \implies dE_h + dE_c = 0 \quad \text{for either sign of } \delta Q.$$

For uniform block temperatures,

$$dS_{\text{universe}} = \delta Q \left( \frac{1}{T_c} - \frac{1}{T_h} \right) \geq 0, \quad T_h > T_c \implies \delta Q \geq 0.$$

The first law conserves energy; the second law selects hot-to-cold heat flow.

## PROBLEM 7

(a) Single phase, fixed composition;  $h, s, v, g$  are mass-specific.

$$\begin{aligned}
 h &= h(T, p), \quad c_p = \left( \frac{\partial h}{\partial T} \right)_p, \\
 dh &= \left( \frac{\partial h}{\partial T} \right)_p dT + \left( \frac{\partial h}{\partial p} \right)_T dp = c_p dT + \left( \frac{\partial h}{\partial p} \right)_T dp. \\
 dh &= T ds + v dp, \quad g = h - Ts, \\
 dg &= dh - T ds - s dT = (T ds + v dp) - T ds - s dT = -s dT + v dp, \\
 \left( \frac{\partial g}{\partial T} \right)_p &= -s, \quad \left( \frac{\partial g}{\partial p} \right)_T = v. \\
 \frac{\partial}{\partial p} \left( \frac{\partial g}{\partial T} \right)_p &= \frac{\partial}{\partial T} \left( \frac{\partial g}{\partial p} \right)_T \implies \left( \frac{\partial s}{\partial p} \right)_T = - \left( \frac{\partial v}{\partial T} \right)_p. \\
 \left( \frac{\partial h}{\partial p} \right)_T &= T \left( \frac{\partial s}{\partial p} \right)_T + v = v - T \left( \frac{\partial v}{\partial T} \right)_p \\
 \implies &\boxed{dh = c_p dT + \left[ v - T \left( \frac{\partial v}{\partial T} \right)_p \right] dp.}
 \end{aligned}$$

(b)  $R$ : mass-specific gas constant.

$$\begin{aligned}
 Z &= \frac{pv}{RT} \implies v = \frac{ZRT}{p}, \quad \left( \frac{\partial v}{\partial T} \right)_p = \frac{R}{p} \left[ Z + T \left( \frac{\partial Z}{\partial T} \right)_p \right]. \\
 v - T \left( \frac{\partial v}{\partial T} \right)_p &= \frac{ZRT}{p} - \frac{RT}{p} \left[ Z + T \left( \frac{\partial Z}{\partial T} \right)_p \right] \\
 &= \frac{ZRT}{p} - \frac{ZRT}{p} - \frac{RT^2}{p} \left( \frac{\partial Z}{\partial T} \right)_p = - \frac{RT^2}{p} \left( \frac{\partial Z}{\partial T} \right)_p. \\
 \implies &\boxed{dh = c_p dT - \frac{RT^2}{p} \left( \frac{\partial Z}{\partial T} \right)_p dp.} \\
 \text{Ideal gas: } Z &= 1 \implies \left( \frac{\partial Z}{\partial T} \right)_p = 0 \implies dh = c_p dT.
 \end{aligned}$$

(c) Molar basis:  $\bar{v}$ ,  $\bar{h}$ ,  $\bar{e}$ , and  $B$ .

$$\begin{aligned}
 R_u &= 8.314462618 \text{ J mol}^{-1} \text{K}^{-1}, \quad \mathcal{M} = 0.0280134 \text{ kg mol}^{-1}, \\
 T &= 150 \text{ K}, \quad dT = 0, \quad \Delta p = (200 - 1) \times 10^5 = 1.99 \times 10^7 \text{ Pa}.
 \end{aligned}$$

Pressure-truncated model:

$$\begin{aligned}
 Z &= 1 + \frac{B(T)p}{R_u T}, \quad \bar{v} = \frac{Z R_u T}{p} = \frac{R_u T}{p} + B(T), \\
 \left( \frac{\partial \bar{v}}{\partial T} \right)_p &= \frac{R_u}{p} + B'(T), \quad B'(T) \equiv \frac{dB}{dT}. \\
 d\bar{h} &= \left[ \frac{R_u T}{p} + B - T \left( \frac{R_u}{p} + B' \right) \right] dp = (B - TB') dp, \\
 \Delta \bar{h} &= \int_{p_1}^{p_2} (B - TB') dp = (B - TB') \Delta p. \\
 \Delta h &= \frac{\Delta \bar{h}}{\mathcal{M}} = \frac{(B - TB') \Delta p}{\mathcal{M}}.
 \end{aligned}$$

(i) **Rigid spheres.**  $N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$ .

$$\begin{aligned} B_{\text{RS}} &= \frac{2\pi}{3} N_A \sigma^3 = \frac{2\pi}{3} (6.02214076 \times 10^{23}) (3.681 \times 10^{-10})^3 \\ &= 6.2908158 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}, \quad B'_{\text{RS}} = 0, \\ \Delta \bar{h}_{\text{RS}} &= (6.2908158 \times 10^{-5}) (1.99 \times 10^7) = 1251.872 \text{ J mol}^{-1}, \\ \Delta h_{\text{RS}} &= \frac{1251.872}{0.0280134} \text{ J kg}^{-1} = \boxed{+44.69 \text{ kJ kg}^{-1}}. \end{aligned}$$

(ii) **Attraction-inclusive model.**

$$\begin{aligned} B - TB' &= [-75.7 - 150(1.01)] \times 10^{-6} = -227.2 \times 10^{-6} \text{ m}^3 \text{ mol}^{-1}, \\ \Delta \bar{h} &= (-227.2 \times 10^{-6}) (1.99 \times 10^7) = -4521.280 \text{ J mol}^{-1}, \\ \Delta h &= \frac{-4521.280}{0.0280134} \text{ J kg}^{-1} = \boxed{-161.4 \text{ kJ kg}^{-1}}. \\ p\bar{v} &= R_u T + Bp, \quad dT = 0 \implies \Delta(p\bar{v}) = B\Delta p, \\ \Delta \bar{e} &= \Delta \bar{h} - \Delta(p\bar{v}) = (B - TB')\Delta p - B\Delta p = -TB'\Delta p. \end{aligned}$$

Rigid spheres:  $B' = 0$ , so only  $p\bar{v}$  raises enthalpy. Attractions lower internal energy; here  $B - TB' < 0$ , reversing the enthalpy change.

$$p = 200 \text{ bar} : \quad Z_{\text{RS}} = 2.0088, \quad Z_{\text{attraction}} = -0.21395.$$

The rigid-sphere correction is not small and  $Z < 0$  is unphysical; these are formal truncated-model results, with model (ii) invalid at 200 bar.

## PROBLEM 8

Steady ideal-gas flow;  $V$ : speed,  $h$ : mass-specific enthalpy,  $M$ : molar mass, bars: molar quantities.

$$q = 0, \quad w_{\text{shaft}} = 0, \quad \Delta s = 0, \quad \Delta z \simeq 0, \quad V_1 \simeq 0.$$

$$T_1 = 1200 \text{ K}, \quad \frac{p_2}{p_1} = \frac{1}{50}, \quad M = 0.0280134 \text{ kg/mol}, \quad R_u = 8.314462618 \text{ J/(mol K)}.$$

(a) **Calorically perfect gas (CPG):**  $\gamma = 1.40$ .

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{(\gamma-1)/\gamma} \implies T_{2,\text{CPG}} = 1200 \left(\frac{1}{50}\right)^{0.4/1.4} = 392.429 \text{ K}.$$

(b) **Thermally perfect gas (TPG):**  $\bar{c}_p(T) = 27.0 + 0.0040T$  in J/(mol K).

$$0 = \int_{T_1}^{T_2} \frac{\bar{c}_p(T)}{T} dT - R_u \ln \frac{p_2}{p_1},$$

$$\int_{1200}^{T_2} \left(\frac{27.0}{T} + 0.0040\right) dT = R_u \ln \frac{1}{50},$$

$$27.0 \ln \frac{T_2}{1200} + 0.0040(T_2 - 1200) = 8.314462618 \ln \frac{1}{50}.$$

MATLAB `fzero` on  $[300, 500]$  K:

$$T_{2,\text{TPG}} = 404.728 \text{ K}, \quad 0 < T_2 < T_1.$$

(c)

$$h_1 - h_2 = \frac{1}{M} \int_{T_2}^{T_1} \bar{c}_p(T) dT, \quad h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \implies V_2 \simeq \sqrt{2(h_1 - h_2)}.$$

CPG:

$$\bar{c}_p = \frac{\gamma R_u}{\gamma - 1} = 29.100619 \text{ J/(mol K)},$$

$$h_1 - h_2 = \frac{\bar{c}_p}{M} (T_1 - T_2) = \frac{29.100619}{0.0280134} (1200 - 392.429) \simeq 838.913 \text{ kJ/kg},$$

$$V_{2,\text{CPG}} = \sqrt{2(838.913 \times 10^3)} \simeq 1295.31 \text{ m/s}.$$

TPG:

$$h_1 - h_2 = \frac{1}{M} \int_{T_2}^{T_1} (27.0 + 0.0040T) dT$$

$$= \frac{27.0(T_1 - T_2) + 0.0020(T_1^2 - T_2^2)}{M}$$

$$= \frac{27.0(1200 - 404.728) + 0.0020(1200^2 - 404.728^2)}{0.0280134}$$

$$\simeq 857.616 \text{ kJ/kg},$$

$$V_{2,\text{TPG}} = \sqrt{2(857.616 \times 10^3)} \simeq 1309.67 \text{ m/s}.$$

(d)

$$\text{CPG : } \bar{c}_p = \text{constant}, \quad \gamma = 1.40,$$

$$\text{TPG : } \bar{c}_p = 27.0 + 0.0040T, \quad \gamma(T) = \frac{\bar{c}_p(T)}{\bar{c}_p(T) - R_u}.$$

The heat-capacity model changes both  $\int \bar{c}_p dT/T$  and  $\int \bar{c}_p dT$ :

$$T_{2,\text{TPG}} = 404.728 > 392.429 = T_{2,\text{CPG}} \quad [\text{K}],$$

$$(h_1 - h_2)_{\text{TPG}} = 857.616 > 838.913 = (h_1 - h_2)_{\text{CPG}} \quad [\text{kJ/kg}]$$

$$\implies V_{2,\text{TPG}} > V_{2,\text{CPG}}.$$

## PROBLEM 11

Ideal, calorically perfect gases; 1: methane, 2: argon, 0: initial state. Neglect partition volume/heat capacity and changes in bulk kinetic/potential energy.  $V = \text{constant}$ ,  $Q_{\text{ext}} = W_{\text{ext}} = 0$ ;  $R_u = 8.3144626 \text{ J}/(\text{mol K})$ .

(a)

$$\begin{aligned} n_1 &= \frac{m_1}{M_1} = \frac{1}{0.0160425} = 62.334424 \text{ mol}, & n_2 &= \frac{1}{0.039948} = 25.032542 \text{ mol}, \\ \bar{c}_{v,i} &= \frac{f_i}{2} R_u, & C_i &= n_i \bar{c}_{v,i} = m_i c_{v,i}, \\ C_1 &= n_1 (3R_u) = 1554.831716 \text{ J/K}, & C_2 &= n_2 \left( \frac{3}{2} R_u \right) = 312.198205 \text{ J/K}. \end{aligned}$$

$$\begin{aligned} \Delta E_1 + \Delta E_2 &= Q_{\text{ext}} - W_{\text{ext}} = 0, \\ C_1(T_{f,a} - 300) + C_2(T_{f,a} - 600) &= 0, \\ T_{f,a} &= \frac{C_1(300) + C_2(600)}{C_1 + C_2} = \boxed{350.164949 \text{ K}}, \\ V_{i,f} = V_{i,0} \implies \Delta S_{i,a} &= C_i \ln \frac{T_{f,a}}{T_{i,0}} + n_i R_u \underbrace{\ln \frac{V_{i,f}}{V_{i,0}}}_0 = C_i \ln \frac{T_{f,a}}{T_{i,0}}. \end{aligned}$$

$$\begin{aligned} \Delta S_{1,a} &= 1554.831716 \ln \frac{350.164949}{300} = \boxed{+240.411 \text{ J/K}}, \\ \Delta S_{2,a} &= 312.198205 \ln \frac{350.164949}{600} = \boxed{-168.127 \text{ J/K}}, \\ \Delta S_a &= \Delta S_{1,a} + \Delta S_{2,a} = \boxed{72.2843 \text{ J/K}} = S_{\text{gen},a} > 0. \end{aligned}$$

Heat transfer across the temperature difference is irreversible.

(b) Same original initial states; ideal gases have  $E_i = E_i(T)$ .

$$C_1(T_{f,b} - 300) + C_2(T_{f,b} - 600) = 0 \implies \boxed{T_{f,b} = T_{f,a} = 350.164949 \text{ K}}.$$

$$\begin{aligned} V_{i,0} &= \frac{n_i R_u T_{i,0}}{p_{i,0}}, \quad p_{i,0} = 10^5 \text{ Pa}, \\ V_{1,0} &= 1.554831716 \text{ m}^3, \quad V_{2,0} = 1.248792821 \text{ m}^3, \\ V &= V_{1,0} + V_{2,0} = 2.803624536 \text{ m}^3, \\ p_f &= \frac{(n_1 + n_2) R_u T_{f,b}}{V} = 90726.52 \text{ Pa} = \boxed{0.907265 \text{ bar}}. \end{aligned}$$

$$V_{i,f} = V, \quad T_{f,b} = T_{f,a} \implies \Delta S_{i,b} = \Delta S_{i,a} + n_i R_u \ln \frac{V}{V_{i,0}}.$$

$$\begin{aligned} \Delta S_{1,b} &= 240.410961 + (62.334424)(8.3144626) \ln \frac{2.803624536}{1.554831716} = \boxed{545.959 \text{ J/K}}, \\ \Delta S_{2,b} &= -168.126641 + (25.032542)(8.3144626) \ln \frac{2.803624536}{1.248792821} = \boxed{+0.197252 \text{ J/K}}, \\ \Delta S_b &= \Delta S_{1,b} + \Delta S_{2,b} = \boxed{546.156 \text{ J/K}} = S_{\text{gen},b} > 0. \end{aligned}$$

(c)

$$\Delta S_{\text{extra}} = \Delta S_b - \Delta S_a \simeq 546.156 - 72.2843 = \boxed{473.872 \text{ J/K}}.$$

Each gas gains access to the whole vessel. I think this is related to ideal-gas mixing, but I have not worked out the connection to the mole-fraction formula.